

Polynomials, Roots, Vieta & Remainders - Teacher Diagnostic Key

Diagnostic routes, hint ladder and verified answers

RECONNECT → DISCOVER → MAKE SENSE → TRY → DIAGNOSE → FADE → ADOPT →
TRANSFER

ALG-03 - Teacher Diagnostic Key

Recognition lab

1. $\alpha + \beta = 7$, $\alpha \beta = 10$; reconstruct target.
2. Factor $(x-5)(x-2)$.
3. $\Delta = 16 - 20 < 0$.
4. Route to ALG-02; complete square $(x-2)^2 + 1$.
5. $x-3$.
6. Evaluate $P(2)$.
7. $P(x-4)$.
8. Each root moves down by 4.
9. Use $x^2 \equiv 1$; remainder 1.
10. $x^3 \equiv 1$; $2026 \bmod 3 = 1$, remainder x .
11. Subtract/combine to eliminate leading terms.
12. $\frac{1}{r} + \frac{1}{s} = \frac{(r+s)}{(rs)}$; need sum/product.
13. Evaluate at $x = -1$; result 0, so yes.
14. $\Delta = 0$.
15. For roots doubled, test a scalar-normalized version of $P(x/2)$.
16. No; a divisor/relation is needed to define reduction.

Practice answers

1. $(x-2)(x-3)$, roots 2,3.
2. sum 7, product 10.
3. 29.
4. $\Delta = 0$; repeated root 3.
5. $P(1) = 8$.
6. sum $5/2$, product $-3/2$.
7. $8^3 - 3128 = 224$.
8. $\Delta = -4$; no real roots.
9. $\Delta = 1$; two distinct real roots.
10. $P(-1) = 0$; yes.
11. $P(x-4) = (x-6)(x-7) = x^2 - 13x + 42$.
12. roots -2, -1.
13. 1.
14. x .
15. $x^5 \equiv 55x - 21 \pmod{x^2 - 3x + 1}$.
16. subtract $\rightarrow -x + 3 = 0$; common root 3.
17. $(\alpha + \beta)/(\alpha \beta) = (-2/3)/(-5/3) = 2/5$.
18. $p^3 + q^3 = s^3 - 3st$.
19. $\Delta = k^2 - 36 = 0$; $k = \pm 6$, repeated roots ± 3 respectively.
20. $P(1) = 0$; remainder 0, factor yes.
21. $41 = 81 - 2\alpha \beta$, so product 20; polynomial $x^2 - 9x + 20$.
22. $x^3 \equiv 1$; exponents 1000,999,998 reduce to 1,0,2 mod 3, so $x + 1 + x^2 \equiv 0$.
23. First at -1: $1 - a + 6 = 0 \rightarrow a = 7$; second $1 - b + 3 = 0 \rightarrow b = 4$. Polynomials roots -1, -6 and -1, -3.
24. $20 \bmod 3 = 2$; reduce $x^2 \equiv -x - 1$, so the canonical remainder is $-x - 1$.

25. roots $(9-1)/2=4$, $(9+1)/2=5$; polynomial $x^2-9x+20$.
26. Vieta avoids computing roots and preserves only requested symmetric information.
27. root count $\rightarrow$ ALG-03 discriminant, no real roots; minimum $\rightarrow$ ALG-02 square completion, min 1.
28. Desired roots **alpha-3, beta-3** come from **$P(x+3)$** .
29. A characteristic relation identifies powers modulo a low-degree polynomial; reduce rather than iterate huge powers.
30. Subtraction/remainder elimination toward a common factor is the elementary form of gcd reasoning.

H0 answers

1. sum **$9/2$** , product **2**; square sum **$81/4-4=65/4$** .
2. **$\Delta=4-60=-56$** ; no real roots.
3. **$\Delta=0$** ; root **$12/(8)=3/2$** .
4. Original roots 2,4; shifted roots 5,7; polynomial **$x^2-12x+35 = P(x-3)$** .
5. Roots move up by 2: if **$P(\alpha)=0$** , **$Q(\alpha+2)=P(\alpha)=0$** .
6. **$3^4+6+7=94$** .
7. **$P(2)=8-12+4=0$** ; yes.
8. **$99 \bmod 3=0$** ; remainder **1**.
9. Relation **$x^2=2x+1$** ; reductions give **$x^6=70x+29$** .
10. subtract $\rightarrow$ **$-x+2=0$** ; common root **2**.
11. sum 10, product 21; **$1000-630=370$** .
12. ALG-02; **$(x-5)^2-4$** , minimum **-4**.
13. **$\Delta=k^2-16>0$** , so **$|k|>4$** .
14. A low-degree relation already identifies the remainder class; direct high-power expansion computes irrelevant information.
15. If old root is **alpha**, **$P((\alpha+5)+5)=P(\alpha+10)$** need not be zero; **$P(x-5)$** is correct for root shift +5.
16. Difference gives a necessary candidate for any common root; original checks establish sufficiency.

Diagnostic tags

- **ALG03-R1** solves roots unnecessarily instead of using invariants.
- **ALG03-R2** discriminant used for an optimization request or vice versa.
- **ALG03-R3** root/input shift sign reversed.
- **ALG03-R4** high powers calculated rather than reduced.
- **ALG03-R5** factor/remainder theorem not connected through evaluation.
- **ALG03-R6** common-root candidate not checked in both originals.